

High-Speed Synchronous Reluctance Machine Rotor Using Multi-Material Additive Manufacturing

Petr Klima , Jan Barta , Daniel Koutny , and Ondrej Vitek 

Abstract—High-speed machines are popular in industry due to their high power density. Permanent synchronous machines are commonly preferred, but they rely on rare earth magnets, which are expensive and environmentally demanding. Axially laminated anisotropic synchronous reluctance machines, a promising alternative, are currently limited to two poles and simple rotor geometry. This paper proposes a modified axially laminated anisotropic geometry, produced using multi-material additive manufacturing, to overcome these limitations in a high-speed synchronous reluctance machine. Magnetic 17-4PH and non-magnetic 316L steels were selected as suitable materials for a case study machine. To determine the effect of heat treatment on the magnetic properties of an additively manufactured 17-4PH steel sample, magnetic measurements were conducted. A 60 000-rpm case study machine was optimized using a non-dominated sorting genetic Algorithm 2, resulting in a machine with a shorter active length. Tests conducted during and after manufacturing verified the feasibility of the proposed rotor solution. This development expands the potential use of axially laminated anisotropic topology and synchronous reluctance machines in high-speed applications. Additionally, the successful use of multi-material additive manufacturing technology in the field of electrical machines is demonstrated.

Index Terms—Axially laminated rotor, high-speed motor, multi-material additive manufacturing, solid rotor, synchronous reluctance motor.

I. INTRODUCTION

MACHINES for high-speed applications attract significant attention in the research community due to their high power density and the potential for direct connection between the machine and the load [1]. Such applications include gas compressors, turbo-molecular pumps, aerospace starter/generators and spindles.

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One of the most commonly used high-speed machines is the permanent magnet synchronous machine (PMSM) [2]. It operates at relatively high speeds and low power, featuring high efficiency and a high power factor. However, permanent magnets are mechanically and thermally vulnerable [3], and their environmentally demanding production, along with their cost, creates pressure to avoid using them.

In applications with low speeds and high power, induction machines (IMs) are utilized [1]. Several IM rotor topologies of are currently in use. A squirrel cage rotor is complex to manufacture, and its mechanical durability is limited [4]. An axially slitted rotor is another option but tends to suffer from lower efficiency [5]. A copper-coated rotor is the most mechanically durable of the IMs due to its purely cylindrical shape, but it is overly complex to manufacture [6].

Another type of machine commonly used in high-speed operation is the switched reluctance machine (SRM) [7]. Its main advantages are a mechanically robust rotor and high-temperature operational capability. However, it often suffers from high torque ripple and lower efficiency compared to PMSMs [8].

While synchronous reluctance machines (SynRMs) are not widely used in high-speed applications, two rotor topologies have been developed to enable their use in such contexts. The first is an air-barrier rotor topology [9], [10], [11], which is popular in low-speed applications but is fragile and requires wide bridges to achieve high-speed operational capability. This reduces its efficiency and severely limits its usability in these applications. The other rotor topology is an axially laminated anisotropic (ALA) rotor [12], [13], [14], [15], which is currently emerging as an alternative to the previously mentioned machine types but is too complex to manufacture. Additionally, only simple barrier shapes can be used, limiting its practical manufacturability to a two-pole rotor. In [17], the authors propose a multi-material additive manufacturing (AM) solution for the ALA topology, but only material characterization and finite element method (FEM) analysis were performed - no rotor was built, and the proposed vision topology is simpler than the solution presented in this paper.

The presented research aims to remove the current restrictions of ALA topology in SynRMs, namely the constraints of a flat or circular shape for non-magnetic material layers and the limitation to two-pole rotors. This could increase the attractiveness of this topology and expand its range of possible applications. The second goal is to introduce the technology of multi-material AM of metals, which the authors patented for ALA topology [16], into the field of electrical machines. To authors' knowledge,

successful implementation and verification of multi-material AM for the rotor of an electrical machine has not yet been presented in scientific publications.

The structure of the paper is as follows. Section II introduces AM, the studied ALA rotor topology, the principle of material selection, and the results of magnetic measurements before and after heat treatment of a selected magnetic material. Section III shows the application of the studied rotor topology in the case study machine. Section IV presents the measurement results of the rotor during manufacturing, and Section V details the test results of the assembled machine, demonstrating the electro-magnetic and mechanical qualities of the case study rotor.

II. AM OF ALA TOPOLOGY

Additive manufacturing of metals is an emerging manufacturing technology that offers new possibilities in the field of electrical machines. It enables the creation of arbitrary structural shapes and material combinations that were previously impractical. However, it is limited by material availability, size restrictions, and high costs. For structural parts of electrical machines, powder-based AM processes, such as selective laser melting (SLM), are particularly attractive due to their manufacturing precision and flexibility [18]. Examples of SLM applications in this area can be found in [19], [20], [21]. Additionally, selective laser sintering is reported in the production of high-silicon content steel laminations [22].

The basic operational principle of this technology is as follows: The first step involves applying a thin layer of powder to a base structure. Desired areas of this powder layer are then melted by a laser beam and welded to the base structure. This process is repeated until the entire object is complete. Non-melted powder serves as support for subsequent layers allowing for the creation of more complex shapes than those achievable with subtractive manufacturing.

From an applications perspective, AM is currently used primarily for the production of windings and stator stacks with special features that enhance the thermal management of machines [23]. Examples of successful AM implementations in the production of stator cores from silicon steel [19], [22] have also been reported. Only a few rotor topologies have been manufactured and tested so far using single-material AM, and these are mostly conventional, such as the SRM rotor presented in [20] and the IM rotor shown in [21], both utilizing high-silicon-content steel. Nonetheless, there is potential to use a multi-material version of this technology to produce rotor topologies with unique qualities not achievable with other manufacturing technologies. Multi-material AM allows the combination of different materials within the same layer into a single structure, with bonding properties similar to those achieved by welding. These attractive properties will be utilized to improve ALA topology, which is the focus of this paper.

A. Rotor Topology

The additively manufactured ALA rotor topology consists of magnetic and non-magnetic materials, similar to the conventional ALA topology as seen in [12] and [14]. However,

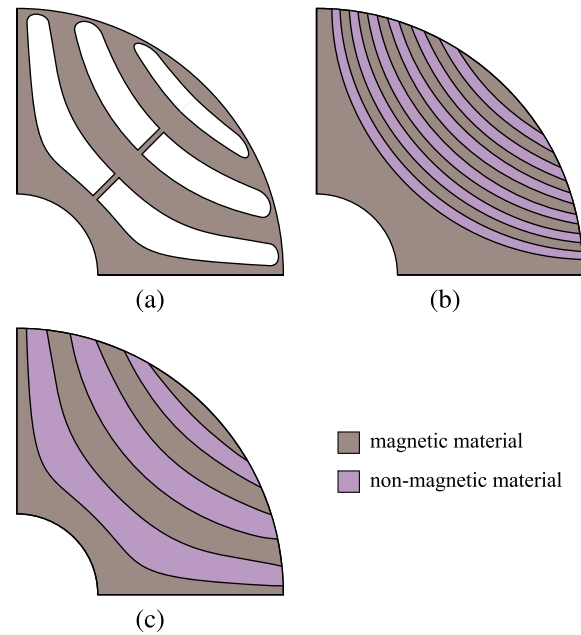


Fig. 1. Examples of (a) air-barrier rotor, (b) conventional ALA rotor, and (c) additively manufactured ALA rotor.

in this case, the design of the non-magnetic layers can follow the same principles as in the air-barrier topology. The number of these layers can follow the recommendations from [24] to reduce torque ripple. The rotors can also have arbitrary shapes and numbers of poles, which is not possible with conventional manufacturing techniques. Examples of air-barrier, classic ALA and proposed ALA rotor topologies are shown in Fig. 1.

As the ALA rotor is intended for high-speed operation, it needs to combine electromagnetic properties with mechanical durability. Because the magnetic and non-magnetic materials making up the rotor structure are bonded together by AM, this creates an effect similar to welding. This results in the rotor exhibiting mechanical properties similar to a homogeneous cylinder, provided that suitable materials are chosen. This can enable very high rotor peripheral speeds, which increase power density. The most important parameters for mechanical properties are density and thermal expansion coefficient. Both materials must have similar values to reduce mechanical stress in the rotor caused by centrifugal forces and increased temperature.

For electromagnetic properties, the permeability of the magnetic material must be as high as possible, while the permeability of the non-magnetic material should be close to that of a vacuum. This results in high saliency, which improves efficiency, torque generation and power factor. Additionally, the electrical resistivity of both materials should be high to reduce eddy-current losses in the rotor.

The use of SLM in the manufacturing process imposes additional demands on the materials used. These include good weldability and availability in the form of gas-atomized powder. The materials should also be commonly used in SLM manufacturing to ensure that suitable manufacturing process parameters are well-established. Additionally, these process parameters should be consistent for both materials to ensure a high-quality product.

TABLE I
MECHANICAL PROPERTIES OF HEAT-TREATED 17-4PH AND
NON-HEAT-TREATED 316L STEEL PRODUCED BY AM [25]

Parameter	Unit	17-4PH	316L
Mass density	kg/m ³	7800	7990
Yield strength	MPa	1116	573
Thermal expansion coefficient	um/m°C	10.8	16.0

TABLE II
LABELS AND HEAT TREATMENT OF SAMPLES USED FOR MAGNETIC
MEASUREMENTS OF 17-4PH STEEL

Label	Heat treatment 1	Heat treatment 2
AB	no heat treatment	
HT-1	1040°C for 2h, air cooling	480°C for 1h, air cooling
HT-2	1040°C for 2h, air cooling	580°C for 4h, air cooling

B. Selected Materials

In line with the previous requirements, the magnetic material 17-4PH (AISI 630, ISO 1.4542) and the non-magnetic material AISI 316L (ISO 1.4404) were chosen for the case study machine. The main reason for this selection is that both are stainless steels with similar manufacturing process parameters and are commonly used in SLM AM.

From a mechanical point of view, the density of both materials is very similar, but there is a difference in the thermal expansion coefficient. This difference can cause mechanical stress in the rotor at elevated temperatures, potentially limiting the maximum operational temperature of the rotor. This needs to be carefully considered to minimize the possibility of catastrophic rotor failure. Table I shows the main mechanical parameters of the chosen materials. The values are valid without heat treatment and may slightly differ after the heat treatments discussed below.

The austenitic 316L should, after heat treatment, have a relative permeability close to 1. Exact magnetic properties of additively manufactured 17-4PH steel are not commonly available. To properly design the case study machine, the magnetic properties of this material were studied and are discussed in Section II-C.

According to the catalogue data [26], the conductivity of 17-4PH and 316L steels at room temperature is 1.25 MS/m and 1.35 MS/m, respectively. These values apply to conventional manufacturing, but a significant change in conductivity is not expected after AM [27].

C. Material Study

AM of the rotor induces residual mechanical stress in the structure. This stress could cause cracks in the rotor even under lower mechanical loads and may degrade the magnetic properties of the materials, so it is necessary to reduce it. This reduction in stress can be achieved through heat treatment. Additionally, heat treatment can be used to harden specific materials, such as 17-4PH steel, to create a martensitic structure with high yield strength.

Three samples of additively manufactured 17-4PH steel were subjected to different heat treatments (see Table II), and their magnetic properties were measured. These heat treatments were proposed by [28] for the combination of 17-4PH and 316L steels.

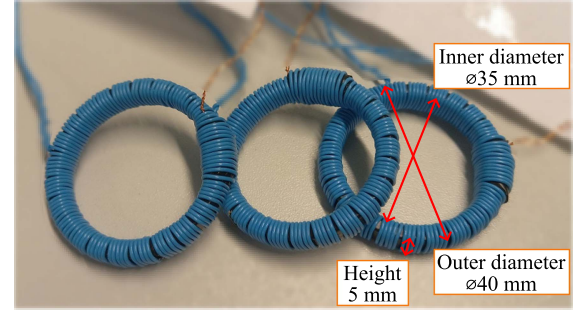


Fig. 2. Ring samples of 17-4PH used for magnetic measurements.

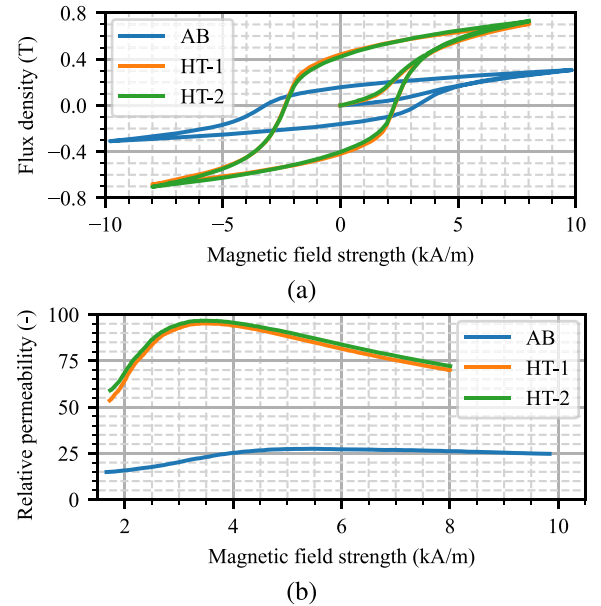


Fig. 3. Measured magnetic properties of AM 17-4PH steel ring samples with different heat treatments. (a) BH curves and (b) relative permeability.

Magnetic measurements were performed using the ring measurement method to test the permeability of the samples and compare the different heat treatments. The samples and their dimensions are shown in Fig. 2.

The measurement results are shown in Fig. 3. They highlight the importance of heat treatment, as the maximum relative permeability is approximately four times higher for both HT-1 and HT-2 compared to AB. HT-2 was selected as the heat treatment method for the use case due to its slightly more beneficial impact on the material's mechanical properties, as found by [28].

The maximum relative permeability of 95 for 17-4PH steel is low compared to steels commonly used in electrical machines, which typically have relative permeability in the thousands. This reduces the achievable saliency, affecting the potential efficiency, power factor and torque of the machine. However, the material is suitable for this study due to its excellent mechanical properties, availability and similar process parameters to those of the non-magnetic steel 316L. Additionally, this material combination has been tested and verified in AM, whereas using a material with higher permeability could result in a rotor with lower quality and a higher risk of structural failure. Moreover, using

TABLE III
CASE STUDY MACHINE MAIN PARAMETERS

Parameter	Unit	Value
Rotor speed	rpm	60 000
Output power	W	2000
Number of poles	-	4
Number of stator slots	-	24
Number of rotor barriers per pole	-	3
Stack length	mm	40
Stator outer diameter	mm	120
Stator inner diameter	mm	62

TABLE IV
OPTIMIZED PARAMETERS AND OBJECTIVES

	Name	Symbol	Unit	Obj.
Parameters	Stator current	I	A	
	Radial air-gap length	δ	mm	
	Barrier height 0/1/2	$H_{b0}/H_{b1}/H_{b2}$	mm	
	Carrier height 0/1/2	$H_{c0}/H_{c1}/H_{c2}$	mm	
Objectives	Torque	T_{avg}	Nm	Max.
	Efficiency	μ	%	Max.
	Torque ripple	T_{ripple}	%	Min.
	Rotor temperature	ϑ_{rot}	K	Min.

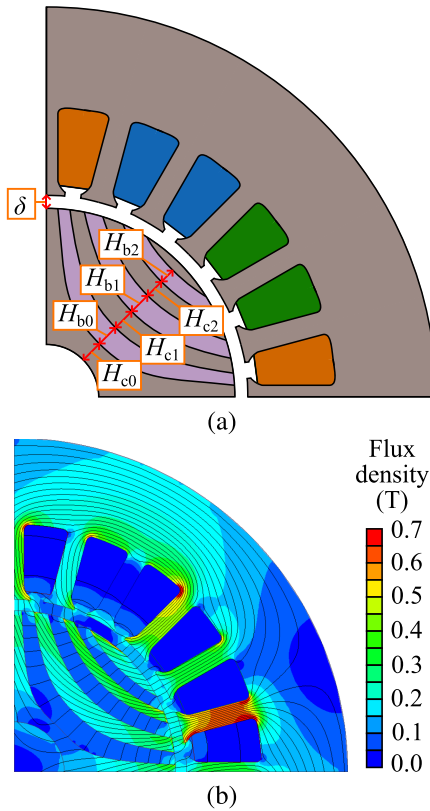


Fig. 4. Case study machine. (a) Electromagnetic model with optimized parameters and (b) flux density distribution in the selected machine at rated operation.

17-4PH steel demonstrate the suitability of multi-material AM ALA topology for high-speed applications, potentially achieving even better electromagnetic parameters with a material of higher maximum permeability in the future.

III. APPLICATION IN CASE STUDY MACHINE

To demonstrate the feasibility of ALA topology manufactured using multi-material AM, a high-speed 60 000-rpm case study machine was designed. The rated speed was chosen to achieve a rotor peripheral speed of 180 m/s. The main parameters of this machine are listed in Table III.

The electromagnetic part of the design process was performed using finite element analysis (FEA). The model used in this analysis is shown in Fig. 4(a). As a trade-off between supply frequency and end-winding length, the machine was designed

with four poles. The rotor barrier geometry was defined according to the natural flux lines shape described in [29]. The number of flux barriers was limited by the resolution of the printer and the subsequent conversion of the smooth geometry to a pixelized form. To achieve continuous flux barriers and reduce the impact of the interface between the two materials, a small number of barriers was desired. This number was chosen based on the recommendations for stator slot numbers presented in [24].

Although a double-layer winding would reduce selected harmonic components and decrease rotor eddy-current losses, a single-layer winding was chosen for the machine due to its higher slot-fill factor and easier manufacturing, given the small size of the stator. The winding was supplied by a sine current set to produce the maximum torque per ampere. Each slot was modelled as a single conductor with a defined current in ampere-turns. The number of turns was determined after optimization, based on to the maximum available supply voltage. The losses identified in the electromagnetic analysis were used as inputs in a lumped parameter thermal network model to calculate the temperatures in the machine. The steady-state lumped parameter thermal network model used consisted of 20 temperature nodes connected by thermal resistances calculated according to [30].

To optimize the machine, a non-dominated sorting genetic algorithm II (NSGA-II) was used. The optimization focused on rotor geometry and flux to achieve a trade-off between torque and efficiency while keeping torque ripple and rotor temperature low. An overview of the optimized machine parameters and the optimization objectives is shown in Table IV. Other model parameters were kept constant.

The optimization results are shown in Fig. 5. The chosen design of the case study machine represents a trade-off among all the optimization objectives. Eddy-current losses in the solid rotor were the main cause of high rotor surface temperatures, which decreased the maximum possible efficiency. These losses were mitigated by a relatively large radial air-gap length of 2 mm, which kept the rotor temperature below 180 °C but resulted in lower saliency and higher stator currents.

The final rotor dimensions of the chosen design are shown in Table V, and the calculated flux density distribution is shown in Fig. 4(b). The low peak value of air-gap flux density is typical for high-speed machines and contributes to reducing rotor eddy-current losses and rotor temperature. This also resulted in a lower power density for the machine, but this was not a concern, as the goal of this case study was to demonstrate the feasibility of AM for high-speed machine rotors.

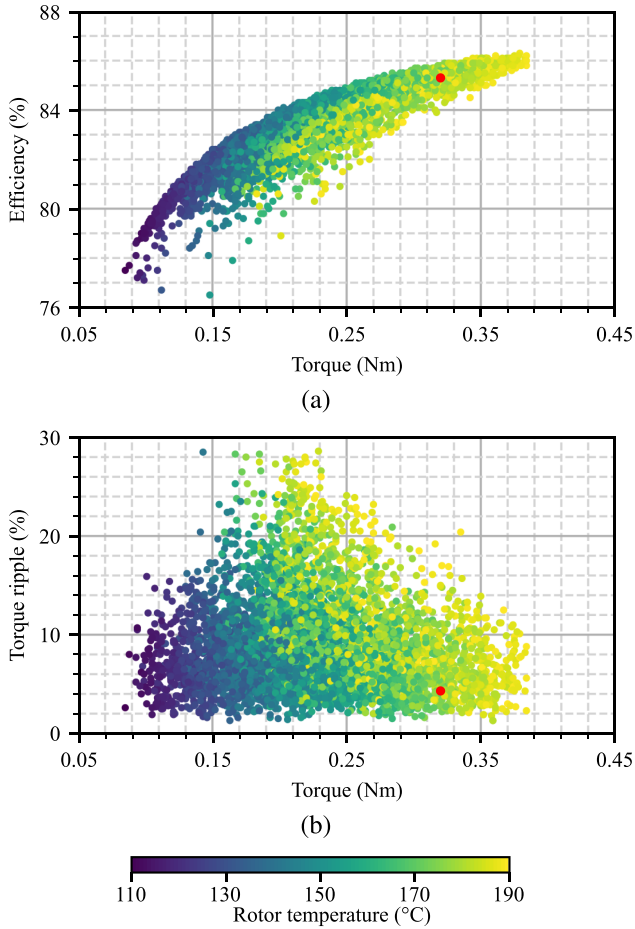


Fig. 5. Optimization results with the chosen design colored red.

TABLE V
FINAL ROTOR DIMENSIONS AFTER OPTIMIZATION

Parameter	Symbol	Unit	Value
Outer diameter	D_{ro}	mm	58
Barrier height 0/1/2	$H_{b0}/H_{b1}/H_{b2}$	mm	3.85/3.25/2.20
Carrier height 0/1/2	$H_{c0}/H_{c1}/H_{c2}$	mm	2.30/3.80/2.05

TABLE VI
LOSSES AND EFFICIENCY OF THE OPTIMIZED CASE STUDY MACHINE

	Unit	Proposed Rotor	Laminated Rotor
Material		17-4PH/316L	M270-35A
Air-gap Length	mm	2.0	2.0
Copper	W	127.5	99.7
Stator Iron	W	78.6	72.8
Losses			
Rotor	W	65.0	5.7
Windage	W	84.4	84.4
Total	W	355.5	262.6
Efficiency	%	85.0	88.4

The calculated efficiency and losses of the proposed additively manufactured rotor are listed in Table VI. To compare the proposed rotor with a conventional laminated rotor, the model was recalculated with the same geometry of flux barriers but using M270-35A lamination. Due to reduced rotor losses from

lamination, a machine with a decreased radial air-gap length was also calculated.

The rotor windage losses were calculated analytically according to [31], while other losses were obtained from electromagnetic FEA. The rotor eddy current losses in the proposed rotor were calculated using a 2D transient analysis in Ansys Maxwell software. The effect of rotor ends on eddy-currents was neglected, as the pole pitch of higher harmonics of magnetic flux is small. To ensure conservative estimates of rotor losses, the conductivity of the rotor materials was not adjusted for increased rotor temperature or rotor end effects. The losses for the laminated rotor were calculated in FEA using the Bertotti loss model for the material.

Due to the solid structure and relatively high conductivity of the materials, the proposed rotor has significantly higher eddy-current losses compared to the laminated rotor. The copper losses are also lower in the laminated rotor due to the higher relative permeability of the lamination, which results in higher d-axis inductance and lower supply current for the same output power. Additionally, the laminated rotor allows for a smaller air-gap, resulting in higher efficiency and potentially greater output power.

In practice, the rotor would require bridges, which were neglected in this calculation. This would slightly increase the copper losses due to higher q-axis inductance. Additionally, the laminated rotor would be mechanically fragile due to the air flux barriers and the lower yield strength of the lamination. Moreover, the solid rotor core produced by AM is more suitable for achieving the highest rotor surface speeds. The bending resistance of the laminated rotor in a classic reluctance machine is lower than that of the solid rotor. Reduced rigidity leads to a decrease in the critical speed, which is unacceptable for high-speed applications [4].

Thus, it can be seen that high rotor eddy-current losses are a significant drawback of the proposed rotor. However, if this rotor proves its functionality, further development could focus on reducing the losses.

After optimization, a static structural analysis of the rotor was performed to evaluate the selected design from a mechanical perspective. Since the rotor is subjected to mechanical stress from rotational centrifugal forces and thermal expansion, the simulation was conducted for three cases: centrifugal forces only, high temperature only and a combination of centrifugal forces and high temperature. The results (Fig. 6) show mechanical stresses below 225MPa, indicating a good safety factor with the selected materials in all tested cases.

IV. VERIFICATION OF ROTOR PROPERTIES DURING MANUFACTURING

To support the previous calculations and design assumptions for the case study machine, a version with a shorter (10-mm) active length was manufactured. This shorter active length was chosen primarily due to the costs of AM and its suitability for verifying the mechanical durability and torque generation capability of the rotor, which could be extended to the full 40-mm active length. Mechanical durability was tested using

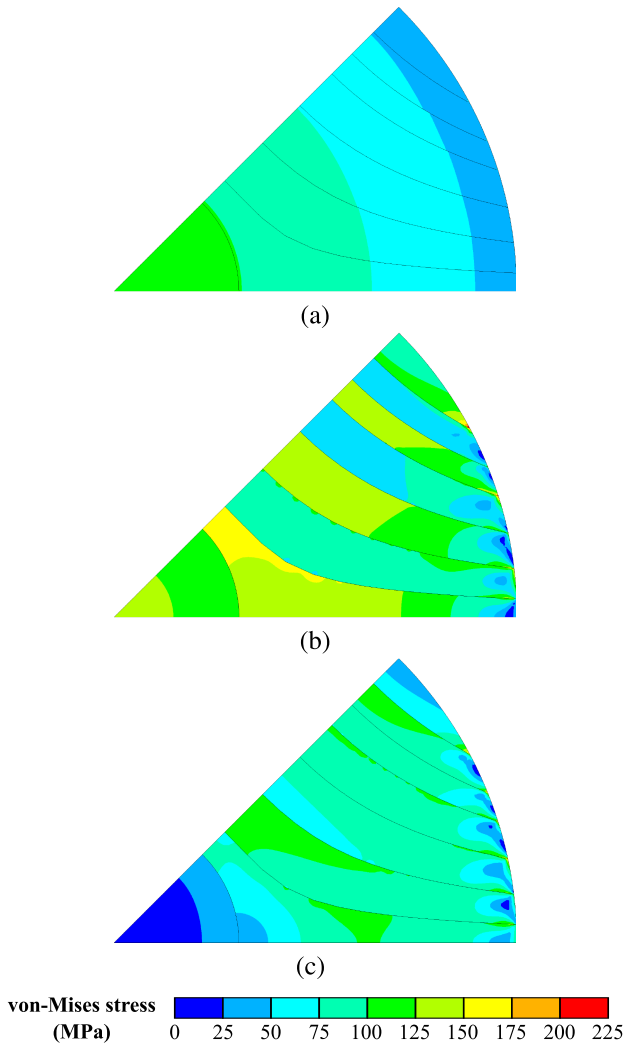


Fig. 6. Von-Mises stress distribution in the rotor at (a) 60 000 rpm, 20 °C; (b) 0 rpm, 200 °C; (c) 60 000 rpm, 200 °C.

microhardness measurements, while torque generation capability was evaluated using inductance measurements. Both tests were conducted as static evaluations before the final machining of the rotor to demonstrate the influence of the selected heat treatment on rotor properties.

To conform with the specifics of the utilized multi-material SLM manufacturing system, the rotor cross-section geometry was converted to a geometry consisting of 0.5 mm × 0.5 mm pixels. This was deemed an appropriate resolution for the rotor geometry, at which the influence on general rotor properties should be marginal. The pixel version of the rotor cross-section is shown in Fig. 7. The rotor was manufactured with increased outer and decreased inner diameters to allow machining to precise dimensions after heat treatment.

The details of the 10-mm rotor before and after heat treatment are shown in Fig. 8. The face of the heat-treated rotor in Fig. 8(b) was polished and etched using acid to highlight the interface between the two materials.

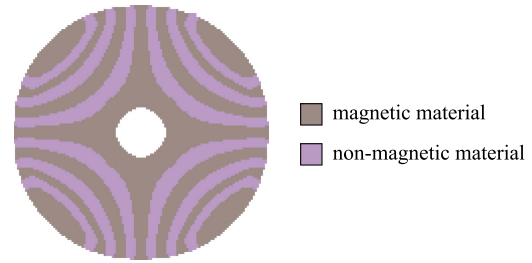


Fig. 7. Pixelized rotor cross-section for manufacturing.

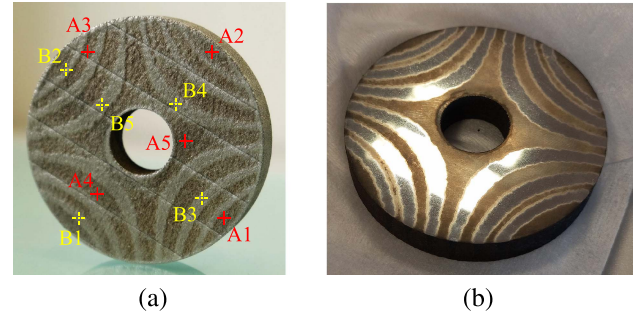


Fig. 8. Manufactured 10-mm rotor, (a) as built, with points of microhardness measurements; (b) after heat treatment, showing the polished and etched face side.

TABLE VII
MEASURED ROTOR HV0.3 MICROHARDNESS AT SELECTED POINTS

Material / Measuring Point	As Built		Heat-treated	
	A-17-4PH	B-316L	A-17-4PH	B-316L
1	254	241	375	192
2	270	243	338	201
3	258	249	357	205
4	262	231	367	203
5	243	246	361	207
Average	257.4	242.0	359.6	201.6
Standard Deviation	10.0	6.9	13.8	5.8

A. Microstructure and Mechanical Properties

To evaluate the materials' state, hardness measurements were performed before and after heat treatment to determine whether martensitic hardening occurred. As shown in Fig. 8(a), the measurement points were distributed across the rotor's cross-section to assess both the main magnetic rotor material (A) and the non-magnetic material (B) in all quadrants, as well as most of the barrier and intra-barrier segments.

The results in Table VII show a significant increase in hardness for material A after heat treatment, and a slight decrease for material B. The heat treatment induced martensitic hardening in material A, which also led to a major increase in inductance, as discussed in Section IV-B. The measured microhardness of material A was approximately 20% lower than the hardness value provided by the manufacturer for the H900 solution in an annealed and aged state [25]. This discrepancy may be due to differences in heat treatment temperatures and dwell times. However, well-accepted tensile strength values can be assumed

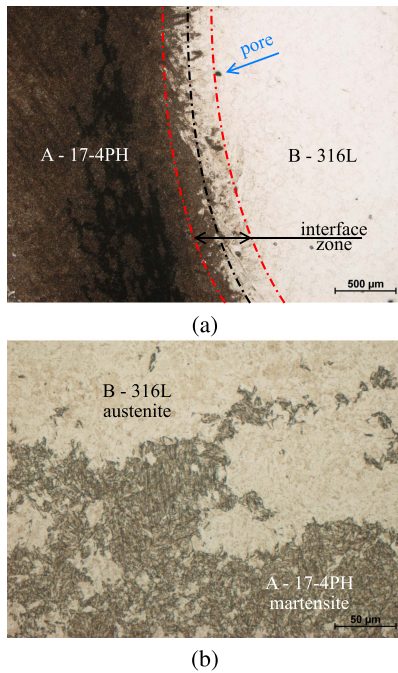


Fig. 9. Magnified interface between magnetic and non-magnetic segments of the rotor cross-section in the direction of the rotor axis: (a) Overview of the interface zone. (b) Detail of austenitic and martensitic microstructure.

(1155 MPa). The hardness of material B decreased by about 16% after heat treatment. Based on the average microhardness measured for material B, a resulting tensile strength of 640 MPa can be assumed, which is similar to the hardness reported in [32] after similar heat treatment. This is around 8% lower than standard non-heat-treated values provided by the manufacturer but still ensures a sufficient safety factor against failure.

A low variance in hardness ($<3.9\%$ for both materials) indicates a uniform microstructure for each material, suggesting that uniform magnetic and electrical conductivity can be expected for the magnetic and non-magnetic materials respectively.

Metallographic preparation and etching of the entire rotor cross-section, as shown in Fig. 8(b), reveal visually distinct areas corresponding to the designed spatial arrangement of the rotor segments. Higher magnification of the interface in Fig. 9 shows the martensitic microstructure of material A and the austenitic microstructure of material B. The interface between individual segments is free of microcracks and excessive porosity. The interface zone is approximately $400\text{--}500\text{ }\mu\text{m}$ wide, which aligns with the expected resolution of a multi-material SLM machine.

B. Inductances

To compare the electromagnetic properties of the rotor before and after heat treatment, an inductance measurement was conducted. A polyamide sleeve was manufactured to hold the rotor inside the stator without influencing the measurement. A picture of the measured stator and rotor is shown in Fig. 10(a).

The stator winding was connected in a star configuration to a 50 Hz voltage source, with the current regulated using a variable resistor. All three phase connections were measured, and the resulting values were averaged. The measurement scheme used

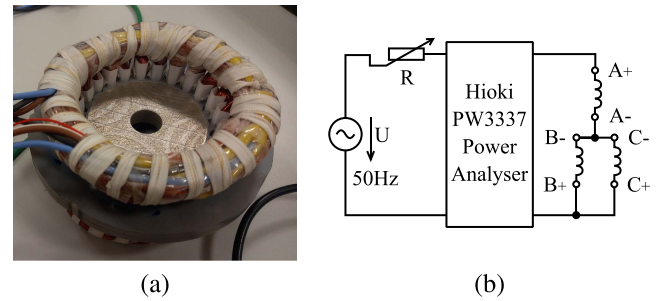


Fig. 10. Inductance measurements. (a) Rotor inserted in the stator. (b) Electrical circuit scheme. A, B, and C are phases of the stator winding.

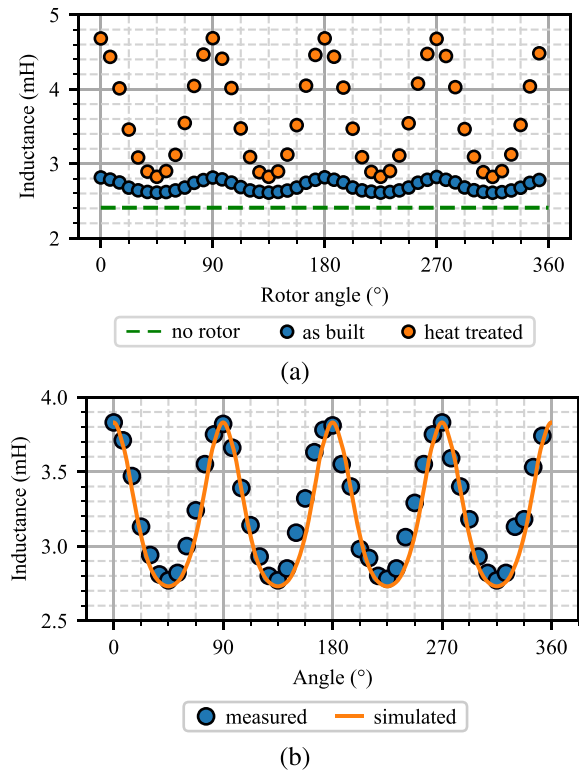


Fig. 11. Measured dependence of inductance on rotor angle at 2 A: (a) Rotor before machining. (b) Final machine.

for standstill verification of synchronous inductance, based on IEEE Standard 1812–2023 [33], is shown in Fig. 10(b).

The total stator-rotor inductance was calculated from the reactive component of the measured power, with the results displayed in Fig. 11(a). The rotor was measured at various angles relative to the stator to assess the inductance's dependence on rotor position. This measurement was conducted both before and after heat treatment, confirming the significance of heat treatment as the peak-to-peak inductance increased by approximately 800%. To ensure a direct comparison, the rotor was measured without a shaft, using post-AM dimensions that differed from the final dimensions. Therefore, the inductance values in the completed rotor are expected to be lower.

The measured inductances confirmed the expected electromagnetic properties of the ALA rotor manufactured through multi-material AM.

TABLE VIII
COMPARISON OF MOTOR INDUCTANCE AND SALIENCY

			Unmachined rotor		Final rotor	
			AB	HT-2	Measured	FEA
Inductance	d-axis	mH	2.85	4.70	3.83	3.83
	q-axis	mH	2.64	2.85	2.77	2.73
Saliency			1.08	1.65	1.38	1.40

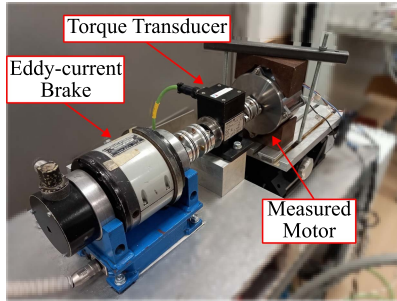


Fig. 12. Motor measurement setup.

V. VERIFICATION OF FINAL MACHINE PROPERTIES

After the heat treatment, the rotor was machined to its final dimensions, and the short (10 mm) version of the case study machine was assembled.

A. Inductance

After the final assembly of the machine, the inductance was measured and compared to the simulated inductance to confirm the validity of the FEA. The results are shown in Fig. 11(b), demonstrating that the measured data closely match the simulated values.

The comparison of final rotor inductance and saliency with the simulated and measured values before rotor machining is presented in Table VIII. Both the simulated inductance and saliency values are close to the measured values. The air-gap of the rotor before machining was approximately 1.05 mm, while the final air-gap was 2.00 mm. This increase in air-gap is the main cause of the drop in saliency after the machine's completion, resulting in a lower saliency ratio for the final machine. Consequently, this leads to a reduction in the expected power density and power factor of the final machine.

B. Torque

To verify the torque production capability of the machine, a static torque test was conducted at maximum torque per ampere operation. The measurement setup for this test is shown in Fig. 12. During the torque measurement, the rotor position was locked at a current angle of 45 degrees, and the motor was supplied with direct current.

The comparison of measured and simulated static torque is shown in Fig. 13. The measured torque deviates from the theoretical curve, indicating that higher supply current is required for higher torque. This discrepancy suggests imprecision in the electromagnetic model, potentially due to variations in the material properties of the magnetic circuit caused by manufacturing.

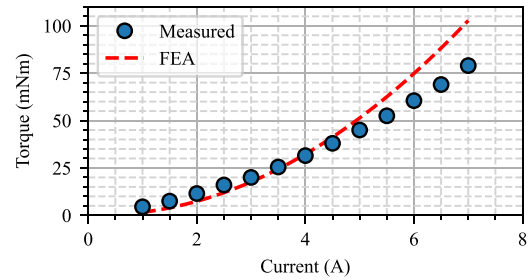


Fig. 13. Comparison of measured and simulated static torque. Current represents direct current supplied to locked-rotor motor.

TABLE IX
COMPARISON OF MEASURED AND SIMULATED MOTOR PARAMETERS AT 15 000 RPM AND RATED TORQUE

Parameter	Unit	Values		Rel. Difference	
		FEA	Measured		
Torque	mNm		82	-	
Output Power	W		128.8	-	
Input Current	A	4.53	4.88	7.7%	
Efficiency	%	63.9	55.2	-13.6%	
Power Factor	-	0.377	0.297	-21.2%	
Loss	Copper	W	67.5	92.9	37.6%
	Stator Iron	W	2.4		
	Rotor	W	1.9	8.8	105%
	Windage	W	1.0	3.0	
	Bearing	W	-		-
	Total	W	72.8	104.7	43.8%

Nonetheless, the torque production capability of the proposed multi-material AM rotor is successfully demonstrated.

C. Losses and Efficiency

Due to the small rated output torque and limitations of the available measuring equipment, the machine was only tested under load up to a maximum speed of 15 000 rpm. The results of this load test at 15 000 rpm, along with the rated torque, are compared to the simulation results in Table IX.

The higher copper losses were primarily due to a lower-than-expected slot-fill factor during manufacturing and higher stator current. Operating at a quarter of the rated speed and having a high ratio of end-winding length to active length resulted in copper losses being the predominant component of the motor's total losses.

The mechanical losses were determined from a run-down test. The combined rotor eddy-current losses and stator iron losses were found to be twice the expected value. This discrepancy could be due to additional losses in the frame or higher-than-anticipated specific iron losses.

The measured efficiency of 55.2% is lower than the rated value, indicating that the full-length case study machine's efficiency is likely to be less than the stated 85.0%.

The maximum speed achieved during no-load operation was 39 000 rpm, corresponding to a rotor peripheral speed of 118 m/s. At this speed, machine vibrations exceeded safety limits, so further increases in speed were avoided to prevent potential failure of the ball bearings.

VI. CONCLUSION

This paper presents the results of magnetic properties measurements for additively manufactured 17-4PH steel. The findings highlight the significant impact of heat treatment on magnetic properties, with the maximum permeability of the material increasing by approximately four times after treatment.

The magnetic 17-4PH steel and non-magnetic 316 L steel were applied in a 2 kW, 60 000 rpm case study high-speed SynRM with a multi-material additively manufactured ALA topology rotor. The design process utilized an NSGA-II optimization algorithm integrated with an electromagnetic FEM model and a lumped parameter thermal network. A static structural FEM analysis assessed the mechanical stress distribution in the rotor, confirming its mechanical robustness with an adequate safety factor.

A version of the case study machine was manufactured with a shortened rotor. Both the mechanical and electromagnetic properties of the additively manufactured rotor were tested before and after heat treatment. Mechanical properties were evaluated through microhardness measurements and material interface analysis, while electromagnetic properties were assessed by inductance measurements of the shaftless rotor inserted in the stator. Following the completion of the machine, tests of inductance, static torque and efficiency were conducted and compared with FEA results. These tests confirmed the rotor's mechanical durability up to two-thirds of the rated speed and torque-producing capability, validating the suitability of the case study machine for high-speed applications and demonstrating the applicability of multi-material AM for high-speed rotors in enhancing ALA SynRM topology.

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